

Assignment 3 Solution

1) 0x ABB9609

= 1010 1011 1011 1001 0110 0000 1001

Fraction bits = $28 - 1 - 6 = 21$

1	010 101	1 1011 1001 0110 0000 1001
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Biased exponent = 010101 = 21

Bias = $2^{6-1} - 1 = 31$

Actual exponent = $21 - 31 = -10$

Fraction = 1 1011 1001 0110 0000 1001

= 0. 1 1011 1001 0110 0000 1001

= 0.8620648384

Decimal = $(-1)^{\text{sign}} \times (1 + \text{fraction}) \times 2^{(\text{exponent})}$

= $(-1)^1 \times (1.8620648384) \times 2^{(-10)}$

= $-1.818422694 \times 10^{-3}$

2) a. $64.2486 * 49.1832$

= $1000000.0011111110 * 110001.0010111011$

= $1.0000000011111110 \times 2^6 * 1.100010010111011 \times 2^5$

= $(1.0000000011111110 * 1.100010010111011) * 2^{11}$

= $1.100010101111111 \times 2^{11}$

3) $28.4810 - (-4.0210)$

= $28.4810 + 4.0210$

= $11100.01111 + 100.00000$

= $1.110001111 \times 2^4 + 1.0000000 \times 2^2$

$$= 1.110001111x2^4 + 0.010000000x2^4$$

$$= 10.000001111x2^4$$

$$= 1.000001111x2^5$$

$$= 1.0000x2^5$$

$$\text{Bias} = 2^{8-1} - 1 = 127$$

$$\text{Biased exponent} = 127 + 5 = 132$$

$$\text{Range of biased exponent} = 0 \text{ to } 2^8 - 1 = 0 \text{ to } 255$$

$$= 1 \text{ to } 254$$

Since 132 is less than upper range(254) and greater than lower range(1), there is no overflow.

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4) feq.d x5, f1, f2
bne x5, x0, jumpEqual
    beq x0,x0, jumpNotEqual
```

```
jumpEqual:
    #code
```

```
jumpNotEqual:
    #code
```